

Strap a Case Bundle SOP

One episode straps one case bundle. It runs from two loose cases standing side by side on a flat strap, an open buckle on the strap's left end, a long tail on its right end and two corner protectors on their spots, to one bundle held by one tight strap, the tail cut short, one corner protector clipped on each top outer corner, and every tool back on its spot.

The order never changes: **wrap the strap, thread the buckle, pull it tight, cut the tail, clip the corner protectors.** Nothing is cut until the strap is tight, and no protector goes on until the tail is cut. The cases are never lifted and never moved apart.

The right gripper does the primary work: it threads the tail through the buckle, pulls the strap tight, cuts the tail, and clips the right corner protector. The left gripper supports: it holds the buckle while the tail goes through, holds the strap flat while it is pulled tight, holds the tail while it is cut, and clips the left corner protector. The one two armed action is the wrap: both grippers lift their own end of the strap and lay it over the top of the cases together, each on its own end. The right arm is the working arm because the rig is set up right handed, so the dominant hand drives it in leader follower. A left handed collection requires a separately approved mirrored variant. Do not swap arms during an episode.

Neither gripper reaches onto the other arm's side to pick anything up or to carry it. Two objects sit on the midline and are shared: the strap, which each gripper meets from its own side, the left from the left end and the right from the right end, and the two cases, which the strap goes around across their full width. Nothing is ever passed from one gripper to the other, only one object is held at a time, and no grasp is changed while an object is off the surface.

Each tool goes back to its own spot before the next step starts. The counts are fixed: one wrap, one thread, one pull, one cut and two corner protectors. There is no second pull and no second cut. An offcut or a protector that lands outside its place is left exactly where it falls, tagged, and the episode continues. Nothing is ever tidied up on camera.

Setup

Complete both checklists before starting an episode.

Cell configuration

- **Environment camera:** 90 mm.
- **cell_type:** bimanual
- **parked_side:** none

Hardware checklist

1. Cameras are on and recording.
2. The environment camera frame holds the whole tabletop in one shot: the strap mat, both cases, both ends of the strap, the cutter spot, the scrap tray and both protector spots.
3. The buckle, the tail and the tops of both cases stay in frame for the whole episode, and the slack in the strap is visible before the pull.
4. Both arms are at home with their grippers open.
5. The tabletop is dry and clear of anything but the items in the Materials checklist.
6. The right arm reaches the tail end of the strap, the buckle from its right side, the top of both cases, the cutter spot, the scrap tray and the right protector spot without extending to a joint limit or folding in on itself. Bring the mat forward until this is true.
7. The left arm reaches the buckle end of the strap, the buckle from its left side, the top of the left case and the left protector spot without extending to a joint limit.
8. Neither arm needs the far side of the table for anything. The paths from each strap end up onto the case tops, from the cutter spot to the buckle, and from each protector spot to its own corner are unobstructed. Draw each arm along its path by hand during setup: it must clear the standing cases on its own side. Widen the gap between the cases and the protector spots until it does.

Materials checklist

1. One flat strap mat lies at the front center of the table, straddling the midline, and is wider than the two cases standing side by side.
2. The mat grips the table. Draw a cloth across it by hand during setup: if it slides, bunches or lifts, weight it or replace it with a rubber mat that stays put. Nothing pins the mat during the episode.
3. One strap lies flat across the mat, straight and with no twist, its middle on the midline, the buckle end running out to the left of the cases and the tail end running out to the right, both ends flat on the table and clear of everything else.
4. The strap is flat webbing, wide enough for a gripper to take it by its face, soft enough to bend over a case corner, and long enough to go around both cases with a tail of at least two hand widths past the buckle.
5. One buckle is fixed on the left end of the strap. Its slot stays open on its own. A buckle that has to be squeezed or held open is not usable: the gripper cannot hold it open and thread it at the same time.
6. Two cases stand side by side on the strap, touching each other, straddling the midline, and the same case is the left case in every episode. The two are clearly different in colour or marking.
7. Both cases are hard sided, flat topped and the same height, so the strap lies flat across both tops in one line. Each has square top outer corners a protector clips onto.

8. Each case is heavy enough to stand still while the strap is pulled tight. A case a pull could drag is weighted or replaced.
9. Two clip on corner protectors lie flat on their own spots, one on the left protector spot and one on the right protector spot, open side up and open side toward their own arm. A protector that has to be squeezed open is not usable: the gripper cannot squeeze it and place it at the same time.
10. One cutter lies flat on the cutter spot, guard down, handle toward the right arm. It cuts the strap through in one closing.
11. One scrap tray stands empty on its spot, front right beside the mat, low enough that the offcut can be dropped into it from above.

Workspace layout

Three rows: the cutter at the back, the mat with the two cases in the middle, the scrap tray a step forward on the right. Everything works in toward the cases.

Back row:

1. Cutter spot, back right, on the right arm's side. Nothing stands at the back left.

Middle row, left to right:

2. Left protector spot, left of the midline, clear of the strap's buckle end.
3. Strap mat, center on the midline, with the two cases standing side by side on the strap. The buckle end runs out to the left and the tail end runs out to the right, both flat on the table.
Shared: each gripper meets the strap from its own side, and both grippers work the case tops.
4. Right protector spot, right of the midline, clear of the strap's tail end.

A clear gap of at least a gripper's width sits between each strap end and the protector spot beyond it, so a protector lifts straight up and travels in to its corner without touching the strap.

Front row:

5. Scrap tray, front right, on the right arm's side, a step forward of the right protector spot and clear of the mat's right edge. Empty at the start, one offcut in it at the end.
6. Clear front strip, the bare table between the mat's near edge and the front table edge.
Nothing stands there at any point in the episode.

No carry passes over a case top before the strap is on it, and no gripper travels to the far side of the table.

Vocabulary

- **Arm assignment:** the right gripper threads, pulls tight, cuts and clips the right protector. The left gripper holds the buckle, holds the strap, holds the tail and clips the left protector. Both grippers lift their own end of the strap and lay it over the case tops together in Step 1. Neither reaches onto the other's side to take or carry anything.
- **Hold:** a closed empty gripper pressed onto an object to keep it still. A hold never lifts, carries or moves what it holds.
- **Tool home:** the spot a tool returns to and lies flat on before the next step starts: the cutter spot. A tool is never left on a case, on the strap, in the scrap tray or in a gripper between steps.
- **Release point:** the gripper opens over a case top when the strap is being laid down, over a corner when a protector is being clipped on, over the scrap tray when the offcut is dropped, or over a tool home. Never over open tabletop.
- **Stable placement:** the object stays where it is put after release, and does not rock, tip, slide or fall off.
- **Case:** one of the two hard boxes that are strapped together. There are two, one on the left of the midline and one on the right, and neither is ever lifted.
- **Case bundle:** the two cases held together as one block by one tight strap. It is the output of the episode.
- **Strap:** the flat length of webbing that goes around both cases. One strap per episode.
- **Buckle:** the slotted buckle fixed on the left end of the strap. Its slot stays open on its own.
- **Buckle end:** the left end of the strap, the end the buckle is on.
- **Tail:** the right end of the strap, the loose end that goes through the buckle.
- **Wrap:** both ends of the strap brought up the outer sides of the cases and laid flat over their tops, so the strap goes all the way around both cases in one line.
- **Twist:** any part of the strap lying on its edge or turned over, so the same face does not run all the way round. A twisted strap is not a wrap.
- **Thread:** the tail pushed through the buckle slot from the right and drawn out the far side, so it comes back over itself.
- **Slack:** any part of the strap that can be lifted off a case. Slack is what the pull takes out.
- **Tight strap:** the strap flat against both case tops and both outer sides, with no slack anywhere and the two cases still touching.
- **Pull:** one continuous draw on the tail, in line with the strap, that takes the slack out. One pull, not a series of tugs.
- **Trim:** one cut across the tail, made once the strap is tight, that leaves a short stub past the buckle.
- **Stub:** the piece of tail left past the buckle after the cut, about a hand's width long. It stays on the bundle.
- **Offcut:** the piece of tail cut off. It goes in the scrap tray and nowhere else.

- **Corner protector:** the clip on cap that goes over a top outer corner of a case, on top of the strap. There are two, one per side.
- **Top outer corner:** the top left corner of the left case and the top right corner of the right case. These are the two corners the strap bends over.
- **Protected corner:** a corner with the strap flat under it and the protector clipped fully over both, sitting square and not hanging off.
- **Quarter turn:** the most any object is tilted in a gripper. Nothing is ever turned over in the air.

Steps

Step 1: Wrap the strap around the two cases

Goal: the strap goes all the way around both cases with no twist, both ends lie flat on the case tops, and the buckle sits on top facing up.

The cases stand on the strap already, so the wrap only brings the two ends up and over. The cases are not lifted and the strap is not pulled out from under them.

1.1 Take one end each

- With the **left gripper**, come down on the buckle end of the strap, a hand's width back from the buckle, and close on its flat face.
- With the **right gripper**, come down on the tail end the same way, a hand's width back from the tip.
- Each gripper takes only its own end. Neither reaches across the cases for the other end.

Check: each gripper holds its own end flat, with the same face of the strap up on both sides. If an end hangs by a corner or slips, lay it back down and take it again. Make no more than two corrective takes.

1.2 Lay both ends over the tops

- Lift both ends together, keeping each one flat, and draw them up the outer side of the case on their own side.
- Turn each end over its own top outer corner and lay it flat along the case top, in line with the part of the strap under the cases.
- Bring the two ends toward each other until they meet and overlap over the join between the cases.
- Keep the strap flat the whole way. Do not let either end roll on its edge or turn over.

Both ends move as one. An end brought up alone drags the strap out from under a case and pulls the other end off its side.

1.3 Set the buckle

- Lay the buckle end down first, so the buckle rests flat on the case tops with its slot facing up.
- Lay the tail down on top of it, so the tail runs across the buckle from the right.
- Open both grippers and lift clear.

Check: the strap runs all the way around both cases in one line with no twist, the buckle lies flat on top with its slot up, the tail lies over it, and both cases are still touching and still on their spots. If the strap is twisted, lay both ends back on the table and wrap again. Make no more than one correction.

Step 2: Thread the tail through the buckle

Goal: the tail is through the buckle slot and drawn out the far side, with the strap still flat around both cases.

2.1 Hold the buckle

- With the **left gripper** closed and empty, come down on the buckle from its left side and press it flat against the case top.
- Take this hold before the tail goes near the slot, and keep it until the tail is out the far side.

The buckle is light and the tail drags as it goes through, so an unheld buckle lifts with the tail or turns over.

2.2 Push the tail through

- With the **right gripper**, take the tail by its flat face, a hand's width back from the tip.
- Bring the tip to the mouth of the slot, low and square to it, and push it straight through from the right.
- Push in line with the strap. Do not lift the tail, twist it, or force it against the buckle frame.

2.3 Draw it out the far side

- Release the tail, take it again on the far side of the buckle, and draw it out until at least two hand widths are through.
- Lay the drawn tail flat along the case top, pointing left, so it runs back over itself.
- The left gripper stays on the buckle the whole time.

Check: the tail is through the slot, comes out the far side, and lies flat and untwisted over the strap. The buckle is still flat with its slot up, and the strap is still around both cases. If the tail has come back out, hold the buckle again and thread it once more. Make no more than one correction.

Step 3: Pull the strap tight

Goal: the strap is tight against both case tops and both outer sides, the two cases still touch, and no slack is left anywhere.

The cases are not held down and are not moved. The pull runs along the top of the cases, not up and away from them.

3.1 Hold the buckle side

- With the **left gripper** closed and empty, keep the buckle pressed flat on the case top.
- This hold does not move again until the pull is finished.

3.2 Pull the tail

- With the **right gripper**, close on the tail beyond the buckle, a hand's width back from the tip.
- Draw it away from the buckle in one continuous pull, low and in line with the case tops, until the strap lies flat everywhere and no slack is left.
- Keep the pull level. Do not lift the tail up off the cases, and do not jerk it.
- Stop when the strap stops giving. Do not pull hard enough to drag a case or dent it.

One pull, and one only. A strap left loose is not pulled a second time and a strap pulled too hard is not let back out. Either one is tagged and the episode continues.

3.3 Let go

- Open the right gripper and lift clear, leaving the tail lying flat along the case top.
- The left gripper releases the buckle and lifts clear only now.

Check: the strap is flat against both case tops and both outer sides, no part of it can be lifted off a case, the two cases still touch, and neither has moved off the mat. If the strap is still loose, leave it and let review flag the episode.

Step 4: Cut the tail

Goal: the tail is cut once, a stub about a hand's width long is left past the buckle, the offcut is in the scrap tray, and the cutter is back on its spot.

Nothing is cut until the strap is tight. The cut is made away from the buckle, never against it.

4.1 Hold the tail and take the cutter

- With the **left gripper**, close on the tail between the buckle and the cut place, and hold it flat and still against the case top.
- With the **right gripper**, close on the cutter handle, lift it straight up off the cutter spot, and carry it to the tail.
- Keep the cutter closed and level on the way over. It never passes over a case top.

4.2 Cut once

- Set the cutter across the tail about a hand's width past the buckle, square to the strap and clear of the buckle itself.
- Stop, then close the cutter once and cut the tail through.
- One cut. Do not saw at the strap and do not cut a second time to tidy the end.

4.3 Drop the offcut and park the cutter

- The left gripper keeps its hold and carries nothing: the offcut stays where it falls if it drops.
- With the **right gripper**, set the cutter down on the case top clear of the strap, take the offcut by its face, carry it to the scrap tray, and open the gripper over the tray.
- Take the cutter again, carry it back to the cutter spot, lay it flat guard down, and release.
- The left gripper releases the stub and lifts clear only now.

Check: the tail is cut through in one clean line, a stub about a hand's width long lies past the buckle, the offcut is in the scrap tray, the strap is still tight, and the cutter lies flat on its spot. If the offcut missed the tray, leave it where it fell and let review flag the episode.

Step 5: Clip on the two corner protectors

Goal: one corner protector clipped square on each top outer corner, over the strap, with the strap still tight.

One protector per carry, the left one first. Each protector is carried by the arm on its own side and goes onto the corner on that same side. Neither passes over a case top on its way.

5.1 Clip the left protector

- With the **left gripper**, close on the protector on the left protector spot, square to it so it hangs open side down.
- Lift it straight up clear of the table and carry it in to the top left corner of the left case, level the whole way.
- Line it up square with the corner, with the strap flat under it, and press it down until it clips on.
- Release and lift clear once it stays on by itself.

5.2 Clip the right protector

- With the **right gripper**, take the protector on the right protector spot the same way.
- Clip it onto the top right corner of the right case, over the strap, and release once it stays on.
- Do not touch the left protector again.

Check: both protectors sit square on their own top outer corner, each with the strap flat underneath it, neither hangs off an edge or sits on the strap's side only, and the strap is still

tight. If a protector sits crooked or comes away, the arm that placed it lifts it clear and clips it on again. Make no more than one correction per protector.

Step 6: End the episode

- Confirm the strap: all the way around both cases, no twist, tight against both tops and both outer sides.
- Confirm the buckle: flat on top, slot up, with the stub through it and lying flat.
- Confirm the cut: one clean cut, the stub about a hand's width long, the offcut in the scrap tray.
- Confirm the protectors: one on each top outer corner, square, over the strap.
- Confirm the cases: still touching, still on the mat, neither lifted or turned.
- Confirm the tools: the cutter flat on its spot, both protector spots empty.
- Correct a failed check only with the correction written in its step. If no correction is written, or the correction would need an earlier step run again, end the episode and reset instead.
- Return both arms home with their grippers open, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unclip both corner protectors by hand and lay them flat back on their own spots, open side up and open side toward their own arm.
2. Pull the stub out of the buckle, lift the used strap off the cases, and put it and the offcut in the bin. A cut strap is never used again.
3. Lay a fresh strap flat across the mat, straight and with no twist, buckle end out to the left and tail end out to the right, both ends flat on the table.
4. Stand the two cases back on the strap, side by side and touching, straddling the midline, the same case on the left as always.
5. Empty the scrap tray into the bin and stand it back on its spot.
6. Pick up any offcut or protector that landed off its place, and wipe the tabletop and the mat clear.
7. Lay the cutter flat on its spot, guard down, handle toward the right arm.
8. Confirm the mat lies flat and still grips the table, neither case is cracked or dented, both protectors still clip on and hold, the buckle slot still stays open on its own, and the cutter still cuts the strap through in one closing.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the start timestamp, violation name, and SOP rule broken. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** the tail goes into the buckle before the strap is all the way around both cases, the pull starts before the tail is out the far side of the buckle, the cutter touches the strap before the pull is finished, a protector goes on before the tail is cut, or a step is skipped.
- **SOP rule broken:** Steps 1 to 5, wrap, thread, pull, cut, clip, in that order, one protector per carry with the left one first.
- **Coaching note:** wrap it, thread it, pull it, cut it, clip it. Nothing is cut on a loose strap and nothing is clipped on a long tail.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes or carries the tail, the cutter, the offcut, or the right protector, or does the thread, the pull or the cut; the right gripper takes or carries the left protector or holds the buckle; or either gripper reaches onto the other arm's side for an object that belongs to that arm.
- **SOP rule broken:** Steps 1 to 5, the right gripper threads, pulls, cuts and clips the right protector, and the left gripper holds the buckle, holds the strap, holds the tail and clips the left protector. The two armed wrap in Step 1 is not a swap: there each gripper works its own end.
- **Coaching note:** if an arm is reaching across, the wrong arm has been chosen. Right works, left holds and owns the left corner.

Violation: Wrap made with one arm

- **Visible cue:** one gripper brings its end up and over while the other end is left lying on the table, or the two ends are brought up one after the other instead of together.
- **SOP rule broken:** Step 1.2, both grippers lift their own end and lay it over the case tops together.

- **Coaching note:** both ends move as one. One end up alone drags the strap out from under a case.

Violation: Strap twisted or off line

- **Visible cue:** the strap lies on its edge or turned over anywhere around the bundle, the top run does not line up with the run under the cases, or the strap sits near a case edge instead of across the middle of the tops.
- **SOP rule broken:** Step 1, the strap goes all the way around both cases in one flat line with no twist.
- **Coaching note:** the same face of the strap runs all the way round. If it is on its edge anywhere, it is not a wrap.

Violation: Strap not all the way around

- **Visible cue:** the strap goes around one case only, an end is left hanging down a side instead of lying on the tops, or the strap is pulled out from under a case during the wrap.
- **SOP rule broken:** Step 1, both ends come up the outer sides and lie flat on both case tops.
- **Coaching note:** both cases inside the strap, both ends on the tops. Check the underside before threading.

Violation: Buckle not held, lifted, or turned over

- **Visible cue:** the left gripper is off the buckle at any point between the tail meeting the slot and the pull finishing; the buckle lifts, slides or turns over as the tail goes through; or the buckle ends up slot down.
- **SOP rule broken:** Steps 2.1 and 3.1, the left gripper holds the buckle flat on the case top from before the thread until the pull is finished.
- **Coaching note:** the holding arm holds through the thread and the pull, not just the thread. The buckle is worked in place.

Violation: Threaded wrong

- **Visible cue:** the tail goes into the slot from the left instead of the right, misses the slot and passes under or beside the buckle, goes in twisted, is forced against the buckle frame, or is left with too little through to grip.
- **SOP rule broken:** Step 2.2, push the tail straight through the slot from the right, flat and in line with the strap, and draw at least two hand widths out the far side.
- **Coaching note:** square to the slot, in from the right, flat all the way. If it will not go, back it out and line it up again.

Violation: Pull done wrong

- **Visible cue:** the tail is lifted up and away from the case tops during the pull, the pull is a

series of short tugs instead of one draw, the strap is jerked, or the pull drags a case off its spot or dents it.

- **SOP rule broken:** Step 3.2, one continuous pull, low and in line with the case tops, stopping when the strap stops giving.
- **Coaching note:** pull along the top, not up off it. One steady draw, then stop.

Violation: Second pull, or the strap let back out

- **Visible cue:** the right gripper pulls the tail again after it has let go, or feeds strap back through the buckle to loosen an over tight strap.
- **SOP rule broken:** Step 3.2, one pull, and one only.
- **Coaching note:** the tightening is one continuous pull. A correction pull makes the episode unlearnable.

Violation: Slack left in the strap

- **Visible cue:** the strap can be lifted off a case top or an outer side after the pull, a gap opens between the two cases, or the bundle comes apart when the grippers let go.
- **SOP rule broken:** Step 3, the strap lies flat against both case tops and both outer sides with no slack anywhere, and the two cases still touch.
- **Coaching note:** watch the strap against the case, not the tail in the gripper. It is tight when it stops giving.

Violation: Cut on a loose strap

- **Visible cue:** the cutter closes on the tail while slack is still visible in the strap, or the cut is made before the grippers have let go of the pull.
- **SOP rule broken:** Step 4, nothing is cut until the strap is tight.
- **Coaching note:** tight first, then cut. A tail cut on a loose strap cannot be pulled again.

Violation: Cut wrong

- **Visible cue:** the cut lands hard against the buckle or leaves a stub far longer than a hand's width, the cutter saws at the strap or closes more than once, the cut runs at an angle across the strap, or the cutter cuts the strap on the wrong side of the buckle.
- **SOP rule broken:** Step 4.2, one square cut about a hand's width past the buckle, clear of the buckle itself.
- **Coaching note:** set it square, a hand's width out, then close once. One cut, no tidying cut.

Violation: Offcut handled wrong

- **Visible cue:** the offcut is left on a case top or the tabletop, is dropped outside the scrap tray and then picked up again, is swept or pushed toward the tray, or the left gripper carries it.
- **SOP rule broken:** Step 4.3, the right gripper carries the offcut to the scrap tray and opens

over it, and an offcut that misses is left where it falls.

- **Coaching note:** carry it to the tray and open above it. A miss is data, the pick up is noise.

Violation: Protector clipped wrong

- **Visible cue:** a protector sits crooked on its corner, hangs off an edge, is clipped on a side or a face instead of a corner, sits on the wrong corner, goes under the strap instead of over it, or comes away after release.
- **SOP rule broken:** Step 5, each protector clips square onto its own top outer corner with the strap flat underneath it.
- **Coaching note:** line it up square with the corner before pressing. Over the strap, never under it.

Violation: Wrong number of protectors

- **Visible cue:** only one protector goes on, both go on the same corner, or a protector is taken off and clipped on again more than once.
- **SOP rule broken:** Step 5, two protectors, one per top outer corner, with no more than one correction each.
- **Coaching note:** the count is fixed at two, one corner each, left first.

Violation: Case lifted, moved, or knocked apart

- **Visible cue:** a gripper lifts, slides, turns or tips a case at any point, the two cases are pulled apart or a gap opens between them, or a case is pushed back into place instead of being left where it is.
- **SOP rule broken:** Steps 1 to 5, the cases stand where they are set and are never lifted or moved.
- **Coaching note:** the strap moves, the cases do not. Nothing in this task requires picking a case up.

Violation: Bad pick, carry, or release

- **Visible cue:** an object moves from one gripper to the other without being set down first, both grippers hold an object at once, a gripper shifts or seats its hold again while the object is off the surface, the cutter or a protector slides along the table with no daylight under it, a protector hangs crooked in the gripper because the grasp was taken off square, or an object is rolled past a quarter turn or turned over in the air.
- **SOP rule broken:** Steps 1 to 5, one object at a time, lifted straight up clear of the surface before any horizontal motion, taken square, tilted no more than a quarter turn, and never passed between grippers.
- **Coaching note:** set it down to change the grip, and lift before the carry. A protector that hangs crooked was taken off square.

Violation: Placement pushed, or correction limit exceeded

- **Visible cue:** a protector, the buckle, the strap, the mat or a tool is shoved, nudged or slid into position instead of being lifted clear and placed again, or a failed check is retried more often than its step allows.
- **SOP rule broken:** Steps 1 to 5, a bad placement is corrected by lifting the object clear and placing it again, and each check allows only the correction it states.
- **Coaching note:** nothing on this station is pushed into line. If grip or approach is forcing repeated correction, let review flag it and move on.

Violation: Tool not returned to its home

- **Visible cue:** the cutter is left on a case top, on the strap, in the scrap tray or on open tabletop; the cutter is set down guard up; or the cutter is still in a gripper when the next step begins.
- **SOP rule broken:** Step 4.3, the cutter returns to the cutter spot and lies flat guard down before the next step starts.
- **Coaching note:** the return is part of the step, not a gap between steps. Lower until it touches, then release.

Violation: Dropped, collided, or knocked over

- **Visible cue:** under command, a gripper or a carried object strikes a case, the buckle, a clipped protector, the scrap tray or the mat and knocks, tips, slides or displaces it; the two arms strike each other over the case tops; or an object falls short of its spot, with no gripper failure, drift or controller fault visible.
- **SOP rule broken:** Steps 1 to 5, keep each object on a clear travel path and release only after a settled placement.
- **Coaching note:** watch the whole path, not just the target. On the buckle, the working arm moves and the holding arm stays put.

Violation: Manipulation outside the camera frame

- **Visible cue:** a wrap, a thread, a pull, a cut or a clip happens partly or wholly outside the environment camera frame.
- **SOP rule broken:** Steps 1 to 5, every manipulation happens inside the camera frame.
- **Coaching note:** the work has drifted outside the framed area. Set the spots back to their setup positions.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with the strap loose, the tail uncut, a protector still on its spot, the cutter out of its home, an arm short of home, or a gripper not fully open.

- **SOP rule broken:** Step 6, confirm the strap, the buckle, the cut, the protectors, the cases and the tools, then return both arms home with grippers open and stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do before the recording ends.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- Recording stopped or paused during the episode (recording system).
- Camera dropped frames or lost its feed (capture system).
- Hardware fault on an arm: gripper failure, drift, controller caused collision, or motor error.
- A spot sits outside an arm's comfortable reach, so the arm stalls at a joint limit or has to fold in on itself to work it. Move the spot and run the Setup checklists again.
- Defective mat: one that slides, bunches or lifts under the pull, or that will not lie flat. Weight or replace it before the next episode.
- Defective strap: one that is frayed, already cut, stiff enough that it will not bend over a case corner, or too short to go around both cases with a tail.
- Defective buckle: one whose slot will not stay open on its own, that will not hold the strap once it is tight, or that lets the strap slip back.
- Defective protector: one whose clip will not stay open on its own, or that will not hold on a corner.
- Defective cutter: one that will not cut the strap through in one closing, or whose guard will not stay down on its spot.
- Defective case: one that is cracked, dented, will not stand flat, or has no square top outer corner for a protector.
- Cases of different heights, so the strap cannot lie flat across both tops. Reset error, not a violation.

Annotation subtasks (from SOP)

1. Take the buckle end of the strap
2. Take the tail end of the strap
3. Lay both strap ends over the case tops
4. Hold the buckle steady
5. Push the tail through the buckle slot
6. Draw the tail out the far side of the buckle
7. Pull the strap tight

8. Take the cutter from its spot
9. Hold the tail steady
10. Cut the tail once
11. Drop the offcut in the scrap tray
12. Return the cutter to its spot
13. Clip one corner protector onto its corner
14. Return both arms home and end the episode